

Navigating Exact-Response Fibres in Shor Order Finding

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An ideal modular multiplier does not uniquely determine its control-level realization. We ask whether this exact-response redundancy can be navigated to improve finite-noise Shor order-finding performance without changing the ideal computation. Across six pre-specified small instances with exact orders 4, 6, 10, and 12, evaluated under a protocol frozen locally before the reported run, we apply intrinsic Euclidean ascent on a four-dimensional exact-response fibre and compare it with an unchanged reference and two fibre-restricted optimizers, basis-invariant random search and SPSA, under the same 5,520-unit composite ledger. The equal-instance-weight held-out advantages over the reference, random search, and SPSA are 4.94×10^{-4} , 3.41×10^{-4} , and 1.20×10^{-4} , with all three hierarchical bootstrap intervals strictly positive. The mean advantage is positive on all six instances while the declared response and the ideal multiplier remain invariant to approximately 10^{-16} . These results are confined to one shared synthetic state-vector implementation and noise family, not to natively compiled circuits, and establish no hardware, fault-tolerance, cryptographic-scale, or asymptotic Shor advantage.

I. INTRODUCTION

Shor’s order-finding routine [11] is built from repeated controlled modular multiplications, and the logical specification of such a block fixes only its ideal input-output response: the permutation $x \mapsto ax \bmod N$. A control-level realization must supply more than that response, and the extra degrees of freedom are not fixed by the specification. Distinct implementations that agree exactly as ideal unitaries can therefore differ in how they couple to a given noise model, and there is no a priori reason for them to deliver the same probability of recovering the exact order. Can implementation freedom that is invisible to the ideal computation remain operationally visible at the noisy task level?

Several literatures bear on this question. Quantum-control landscape theory established that optimal controls are generically non-unique and that level sets of a control objective form connected families [1, 2], and observable-preserving homotopy methods such as D-MORPH make motion along those level sets constructive [3, 4]. Gradient-based pulse optimization [5], composite pulses [6], dynamically corrected gates [7], and robust control design [8] all exploit residual control freedom to suppress errors, but they target gate-level or ensemble-level fidelity. On the compilation side, equivalence-preserving circuit optimization [9] and noise-adaptive compilation [10] choose among implementations of a fixed logical circuit using cost models or device calibration data rather than the success probability of the algorithm being compiled.

The construction below differs on three specific axes: the invariant is the full declared ideal multiplier rather than a scalar objective or a gate fidelity; the objective is downstream exact-order recovery evaluated on held-out noise realizations rather than a gate metric; and the comparison is made across a frozen multi-instance family under a matched evaluation budget. To our knowledge, we have not identified a prior study that combines exact ideal-multiplier preservation, fibre-restricted naviga-

tion, held-out exact-order recovery, and a frozen multi-instance comparison. We make no priority claim beyond that conjunction of features.

We work with a frozen nine-parameter synthetic implementation ansatz whose declared response is a linear map $R \in \mathbb{R}^{6 \times 9}$ of rank 5. The exact ideal-response fibre is $\mathcal{F}_0 = \{\theta \in \mathbb{R}^9 : R\theta = 0\}$, of dimension 4. Within the declared ansatz, $R\theta = 0$ preserves the declared response and, through the explicit implementation map of Sec. II, also implies $U(\theta) = U_0$ exactly. Choosing an orthonormal basis T of $\ker R$ gives the intrinsic parameterization $\theta = Tz$ with $z \in \mathbb{R}^4$, so that every iterate satisfies $R\theta = RTz = 0$ by construction. We maximize the training task success in the coordinates z and compare the resulting implementation with an unchanged reference and with two optimizers restricted to the same fibre, basis-invariant random search and SPSA, under the same 5,520-unit composite ledger. Equality is equality of the total charged ledger, not of each call category separately; the decomposition is given in Sec. III.

Under this frozen protocol the intrinsic method improves shifted held-out exact-order recovery on all six instances, whose exact orders are 4, 4, 6, 6, 10, and 12. The equal-instance-weight mean advantages are 4.94×10^{-4} over the reference, 3.41×10^{-4} over random search, and 1.20×10^{-4} over SPSA, and all three hierarchical bootstrap intervals lie strictly above zero. Each of the three comparisons has a positive instance-level mean on 6/6 instances, and the declared response and ideal multiplier are preserved to approximately 10^{-16} throughout. The advance over an earlier single-instance preflight is the directional consistency of these signs rather than a larger effect size; the absolute gains remain of order 10^{-4} in success probability.

Two claims follow, and no more. Conceptually, exact ideal-response redundancy is not an operationally inert implementation degeneracy in this model: the ideal computation is held fixed while the noisy task score moves. Empirically, a discrete intrinsic navigation of the fibre improves held-out task performance across one frozen

six-instance synthetic family. The construction is purely finite-dimensional and self-contained — the declared response is a linear map on the implementation parameters, its kernel is the fibre, and the search is an ordinary gradient ascent in an orthonormal basis of that kernel — and the trajectory is a discrete, uncertified finite-difference scheme rather than a certified continuous dynamics. The ansatz is synthetic rather than a native gate decomposition, the noise family is synthetic and shared across instances, and nothing here bears on hardware performance, fault tolerance, cryptographic scale, or asymptotic Shor complexity.

II. EXACT-RESPONSE FIBRE AND TASK OBJECTIVE

A. Synthetic implementation ansatz

For an instance (N, a) with $\gcd(a, N) = 1$ we study the modular-multiplication permutation $x \mapsto ax \bmod N$, realized by the ideal multiplier U_0 on a work register of dimension $d_N = 2^{\lceil \log_2 N \rceil}$, with $U_0|x\rangle = |ax \bmod N\rangle$ for $x < N$ and $U_0|x\rangle = |x\rangle$ for $N \leq x < d_N$. The six instances use $d_N = 16, 16, 32, 32, 64, 64$ respectively. Around U_0 we introduce a nine-parameter synthetic implementation ansatz whose declared response is the linear map $R \in \mathbb{R}^{6 \times 9}$ of Appendix A. The serialized response matrix has six rows, five of which are independent; the sixth is the sum of rows 1, 2, 3, and 5 and is retained as an explicit consistency/audit coordinate. The same R is used for every instance.

Writing $q = R_{1:5} \theta$ for the five independent response coordinates, the ansatz is

$$U(\theta) = U_0 D(q),$$

$$D(q) = \text{diag}_n \exp \left[-\frac{i}{2} \sum_{k=1}^5 q_k g_k(n) \right],$$

where n indexes the work-register basis, $w_j(n)$ is the j -th binary digit of n , and the five commuting generators are

$$g_k(n) = (-1)^{w_{k-1}(n)}, \quad k = 1, \dots, 4,$$

$$g_5(n) = (-1)^{w_0(n) + w_1(n)}.$$

The generators depend only on the four lowest work-register digits for every instance. This is a design choice of the synthetic ansatz, made so that one fixed R and one fixed $D(\cdot)$ apply across instances of different register size; the five generators are not claimed to span the control degrees of freedom of a native modular-multiplier decomposition, and the ansatz is not such a decomposition.

B. Exact-response invariance

Since $\text{rank}(R) = 5$ and $\theta \in \mathbb{R}^9$, the kernel of R has dimension 4. The exact-response fibre is

$$\mathcal{F}_0 = \{\theta \in \mathbb{R}^9 : R\theta = 0\}, \quad \dim \mathcal{F}_0 = 4.$$

The tangent space of $\mathcal{F}_0$ is the range of the orthogonal projector $P_T = I_9 - R^\dagger R$, with $R^\dagger$ the Moore-Penrose pseudoinverse. We fix an orthonormal basis $T \in \mathbb{R}^{9 \times 4}$, $T^\top T = I_4$, so that every $\theta \in \mathcal{F}_0$ is $\theta = Tz$ with $z \in \mathbb{R}^4$.

Because $D(0) = I$, we have

$$R\theta = 0 \Rightarrow q = 0 \Rightarrow D(q) = I \Rightarrow U(\theta) = U_0,$$

and in particular every trajectory $\theta(s) = Tz(s)$ satisfies $R\theta(s) = RTz(s) = 0$ and implements the same ideal modular multiplier as the reference realization. Three statements must be kept distinct throughout: $R\theta = 0$ is declared-response invariance; $U(\theta) = U_0$ is the identity of the ideal multiplier within the declared ansatz, which follows from the previous line; and the residuals of order 10^{-16} reported in Sec. V are the floating-point verification obtained by executing the code. The third is a numerical check at machine precision, not a formal certification.

C. Exact-order success metric

For an instance (N, a) let $r = \text{ord}_N(a)$. The quantum phase estimation (QPE) routine [12, 13] uses m control qubits with $M = 2^m$, where m is fixed by the frozen selection rule of Sec. III D. Under the synthetic noise model of Appendix B, the ideal (non-dephased) outcome distribution for a fixed noise realization ξ and phase offset vector $\phi_x(\theta, \xi)$ is

$$q(y; \theta, \xi) = \frac{1}{r} \sum_{s=0}^{r-1} |A_s(y; \theta, \xi)|^2,$$

with amplitudes

$$A_s(y; \theta, \xi) = \frac{1}{M} \sum_{x=0}^{M-1} \exp \left[2\pi i x \left(\frac{s}{r} - \frac{y}{M} \right) + i\phi_x(\theta, \xi) \right].$$

The prefactor $1/M$ is the product of the state normalization $M^{-1/2}$ and the unitary inverse-QFT normalization $M^{-1/2}$, so that $\sum_y |A_s(y)|^2 = 1$ for every s and hence $\sum_y q(y; \theta, \xi) = 1$ before dephasing. After applying dephasing with rate η_ξ , the measured distribution is

$$p(y; \theta, \xi) = (1 - \eta_\xi) q(y; \theta, \xi) + \frac{\eta_\xi}{M},$$

which is already normalized, so the residual renormalization $p \leftarrow p / \sum_y p(y)$ applied in the code is a numerical no-op and the dephasing weight of the mixture is exactly η_ξ .

The success probability is

$$S(\theta; \xi) = \sum_{y \in \mathcal{B}} p(y; \theta, \xi),$$

where

$$\mathcal{B} = \{y : \text{den}[\text{CF}_{\leq N}(y/M)] = r\}$$

is the frozen set of measurement outcomes whose continued-fraction denominator equals the exact order r . The rule is implemented as follows. For each outcome y the exact rational y/M is replaced by the closest rational whose denominator is at most N ; this is a best rational approximation subject to the denominator bound, so semiconvergents are admitted and not only the last convergent. The outcome enters $\mathcal{B}$ if and only if the denominator of that approximation, in lowest terms, equals r exactly. The bound is N , never the known order r . When two candidates are equidistant the implementation returns the one with the smaller denominator. A best approximation always exists, so no outcome is undefined; $y = 0$ maps to $0/1$ and never enters $\mathcal{B}$. Only exact-order recovery counts as success; no classical retry, no multiple of r , and no post-selection is credited.

This is the point at which the two levels separate. The map $\theta \mapsto \phi_x(\theta, \xi)$ of Appendix B depends on the full nine-parameter θ , not on $q = R_{1.5}\theta$ alone. Equality of the ideal modular multiplier therefore does not imply equality of the finite-noise order-finding distribution: two implementations with $R\theta = 0$ are identical as ideal unitaries and may still differ in $S(\theta; \xi)$.

This separation is a stipulated property of the synthetic noise family, not an observed one. The ideal multiplier depends only on $R_{1.5}\theta$, whereas the finite-noise phase map depends on the full parameter vector through $C_\xi\theta$. The experiment therefore does not test whether such a separation arises in a native circuit or a physical device. It tests whether, once present in the declared model, the resulting exact-response freedom defines a navigable task landscape with held-out predictive value. Given this declared separation, the remainder of the paper tests whether the residual task dependence can be navigated in a way that transfers from training seeds to shifted held-out seeds.

For each meta-repetition we evaluate

$$\hat{S}_{\text{train}}(\theta) = \frac{1}{N_{\text{train}}} \sum_{j=1}^{N_{\text{train}}} S(\theta; \xi_j)$$

on a training set of noise seeds. Held-out performance $\hat{S}_{\text{held}}(\theta)$ is evaluated on a separate, distribution-shifted seed set (Appendix B) and is used for all reported comparisons.

III. PROSPECTIVELY FROZEN METHODS AND PROTOCOL

A. Intrinsic Euclidean ascent

The primary method is a discrete Euclidean gradient-ascent scheme in the intrinsic fibre coordinates z . It is a finite-dimensional discrete navigation motivated by the geometric-flow viewpoint; no continuous dynamics is integrated and no certification of a trajectory is claimed. We compute the finite-difference gradient

$$g(z) = \nabla_z \hat{S}_{\text{train}}(Tz),$$

and update

$$z_{k+1} = z_k + \alpha_k \frac{g(z_k)}{\|g(z_k)\|},$$

with $\alpha_k = \min(\rho, \|g(z_k)\|)$ and trust radius $\rho = 0.10$.

The executed rule is the following. The gradient is estimated by a central difference with step $h_{\text{grad}} = 2 \times 10^{-5}$. If $\|g\| < 10^{-14}$ the iteration terminates. At each iteration the line search evaluates the current point and then tests $\alpha = 1$, followed when necessary by $\alpha = 1/2$; the first candidate that strictly improves the training objective is accepted, ties retain the earlier candidate, and if neither candidate improves the objective the current iterate is retained. Iteration stops when the remaining composite ledger cannot fund another complete iteration, and any residual units are spent on response-only audit calls.

The intrinsic and SPSA methods return their final iterates. Basis-invariant random search returns whichever of the reference point and the sampled candidates attains the largest training score, with ties retaining the earlier candidate.

Since $\theta_{k+1} = Tz_{k+1}$ and $RT = 0$, the identity $R\theta_{k+1} = 0$ holds at every iterate, so preservation of the declared response is a property of the parameterization rather than a numerical result checked afterwards.

B. Equal-budget comparators

Basis-invariant random search. We sample uniformly in the radius-1.6 Euclidean four-ball embedded in the fibre:

$$v \sim \mathcal{N}(0, I_9), \quad u = P_T v, \quad \theta = 1.6 U^{1/4} \frac{u}{\|u\|},$$

with $U \sim \text{Unif}(0, 1)$. The exponent $1/4$ gives the uniform volume distribution in the four-dimensional ball, which makes the comparison invariant under rotations of the fibre basis.

SPSA. We use the standard two-sided gradient estimator on the loss $L = 1 - \hat{S}_{\text{train}}$ [14]. In the tangent coordinates,

$$\hat{g}_k = \frac{L(T(z_k + c\Delta_k)) - L(T(z_k - c\Delta_k))}{2c} \Delta_k,$$

where Δ_k has i.i.d. Rademacher entries and $c = 0.03$, with update

$$z_{k+1} = z_k - \eta_k \hat{g}_k, \quad \eta_k = \frac{0.18}{1 + 0.02k}.$$

Reference. The unchanged implementation $\theta = 0$. It consumes no optimization budget and serves solely as the fixed no-navigation control.

Ambient motion off the fibre is not an admissible comparator, because it can alter the declared response. Both optimizing comparators are therefore restricted to the same fibre as the intrinsic method.

The five free constants of the three methods are the intrinsic trust radius $\rho = 0.10$, the intrinsic gradient difference step $h_{\text{grad}} = 2 \times 10^{-5}$, the random-search radius $r_{\text{max}} = 1.6$, the SPSA perturbation $c = 0.03$, and the SPSA gain schedule $\eta_k = 0.18/(1 + 0.02k)$. These values form part of the frozen v0.9.4 protocol and are unchanged from the archived v0.9.3 preflight driver. The comparisons below establish performance only at these stated hyperparameters and do not establish optimizer-wide dominance; no sweep or sensitivity analysis over them was performed.

C. Composite evaluation ledger

The three optimizers – intrinsic, basis-invariant random, and SPSA – receive identical composite ledgers of 5,520 charged evaluation units per meta-repetition. The unchanged reference performs no optimization and is evaluated only as a fixed control; it is not an equal-budget optimizer. A charged unit is either

- a task call: evaluating the success probability for one noise seed (a training mean over five seeds costs five task calls);
- a response call: evaluating $R\theta$.

The composite ledger is the sum of the two; all three optimizers terminate at exactly 5,520 charged units. Held-out evaluation is performed only after selection and is not charged. A method may spend the remainder of its budget on response calls, which carry no optimization information and are included only to enforce equal accounting.

Table I gives the decomposition, which is identical in all 72 instance/meta-repetition records. Equality is equality of the total charged ledger, not of each call category separately: the intrinsic method spends 20 of its units on response padding and therefore makes 20 fewer task calls than either comparator. The comparison consequently does not by itself establish equality of wall-clock cost or of task-information access.

TABLE I. Composite ledger decomposition, identical across all six instances and all 12 meta-repetitions each.

Optimizer	Task calls	Response/padding	Composite total
Intrinsic	5500	20	5520
Random	5520	0	5520
SPSA	5520	0	5520

D. Frozen multi-instance design

The following design was frozen before the reported full run:

- Instances: $(N, a) \in \{(15, 2), (15, 7), (21, 2), (21, 5), (33, 2), (35, 2)\}$, with exact orders 4, 4, 6, 6, 10, 12.
- For each instance, m is the smallest value in $[\lceil \log_2 N \rceil, 2\lceil \log_2 N \rceil]$ for which the frozen continued-fraction success set $\mathcal{B}$ is nonempty.
- For each instance: 12 meta-repetitions, 5 training seeds per meta-repetition, 8 shifted held-out seeds.
- Composite budget of 5,520 charged units per optimizer per meta-repetition.
- Initial point $z = 0$ for the intrinsic method and for SPSA.

All instances executed as part of the frozen v0.9.4 full-run protocol are reported here. Earlier development runs and the v0.9.3 preflight lie outside that protocol and contribute no data to the reported statistics.

E. Statistical inference and success criteria

For each instance we run $N_{\text{meta}} = 12$ meta-repetitions and record, on held-out seeds,

$$\Delta_X = \hat{S}_{\text{held}}(\theta_{\text{intrinsic}}) - \hat{S}_{\text{held}}(\theta_X),$$

for $X \in \{\text{reference}, \text{random}, \text{SPSA}\}$. All comparator differences are paired within instance and meta-repetition: the three optimizers see the same five training seeds, the final candidates are evaluated on exactly the same eight held-out seeds, and the bootstrap resamples these paired records intact.

The primary inference is an equal-instance-weight hierarchical bootstrap: instances are resampled with replacement, then meta-repetitions are resampled with replacement within each sampled instance, and the equal-weight mean over sampled instances is recorded. We report the point estimate, the 95% percentile interval over 10,000 draws, and the number of instances with positive mean advantage. Equal instance weighting prevents any single instance from dominating the aggregate through a larger effect size.

The frozen primary success criteria are:

1. aggregate 95% interval for intrinsic – reference strictly above zero;
2. aggregate 95% interval for intrinsic – random strictly above zero;
3. aggregate 95% interval for intrinsic – SPSA strictly above zero;
4. intrinsic beats random on at least 2/3 of instances;
5. intrinsic beats SPSA on at least 2/3 of instances;
6. all frozen structure, budget, and exact-fibre gates pass.

IV. MOTIVATING SINGLE-INSTANCE PREFLIGHT

An earlier frozen single-instance preflight on $(N, a) = (15, 2)$ motivated the present design but contributes no data to the claims below. The independently seeded protocol reported here repeats that instance under a new stable seed-derivation rule — so its $(15, 2)$ numbers are a fresh same-instance repetition rather than a bitwise reproduction — and adds five instances spanning orders 4, 6, 10, and 12. Its principal purpose is to test cross-instance directional consistency under a protocol frozen before the reported run. The preflight bundle is archived separately and its numerical summary is recorded there.

V. MULTI-INSTANCE RESULTS

A. Aggregate advantage

Under the frozen six-instance protocol, intrinsic exact-fibre navigation achieves higher held-out performance than the three specified comparators — the unchanged reference, the equal-budget basis-invariant random search, and the equal-budget SPSA configuration — while preserving the ideal multiplier to numerical precision. Table II reports the equal-instance-weight aggregate. All three intervals lie strictly above zero, so criteria 1–3 pass; criteria 4 and 5 pass at 6/6 rather than the required 4/6.

TABLE II. Equal-instance-weight aggregate advantage of the intrinsic method over each comparator. Intervals are equal-instance-weight hierarchical bootstrap 95% percentile intervals that resample instances and then meta-repetitions within instances.

Comparator	Mean	95% interval	Pos. inst.
Reference	4.94×10^{-4}	$[2.90, 6.82] \times 10^{-4}$	6/6
Random	3.41×10^{-4}	$[2.35, 4.55] \times 10^{-4}$	6/6
SPSA	1.20×10^{-4}	$[0.60, 1.93] \times 10^{-4}$	6/6

These intervals quantify variation under the frozen resampling scheme; they do not quantify implementation-family or hardware-model uncertainty. The three comparisons are unadjusted co-primary intervals under the decision rule frozen before the reported run, and no family-wise error guarantee is claimed. The hierarchy contains only six top-level instances, so the percentile intervals should be interpreted cautiously: they quantify resampling variation within this frozen family and do not establish coverage over a broader population of order-finding instances, or over alternative implementation and noise families.

As a complementary descriptive check, all six instance means have the same sign for each comparator, corresponding to a two-sided exact sign-test value of $2(1/2)^6 = 0.03125$. This calculation was not part of the frozen primary decision rule and is reported only as a post hoc directional summary.

B. Per-instance directional consistency

An aggregate can be driven by a single strong instance, so we report the per-instance means in Table III. The sign of the advantage is the same on every instance and against every comparator: 18/18 instance-comparator cells are positive. The magnitudes vary by roughly a factor of four across instances, and the margin over SPSA is the smallest of the three throughout, falling to 0.55×10^{-4} on $(35, 2)$. The gains are statistically directionally consistent within the frozen family but small in absolute success-probability terms; the experiment establishes navigability rather than practical computational advantage.

TABLE III. Per-instance mean held-out advantage of the intrinsic method ($\times 10^{-4}$), each averaged over 12 meta-repetitions. Here $r = \text{ord}_N(a)$ and m is the frozen number of control qubits.

Instance	r	m	vs. reference	vs. random	vs. SPSA
(15, 2)	4	4	6.96	3.57	2.03
(15, 7)	4	4	7.64	2.92	0.68
(21, 2)	6	7	1.94	2.31	0.61
(21, 5)	6	7	7.30	5.77	2.70
(33, 2)	10	7	1.76	1.85	0.63
(35, 2)	12	8	4.03	4.05	0.55

Table IV gives the absolute scale against which those differences should be read. Absolute exact-order probabilities differ across instances because the frozen control-register sizes and exact-order success sets differ. The aggregate of Table II therefore averages absolute success-probability differences with equal instance weight; it is not a normalized relative-gain statistic. The corresponding absolute levels for the two optimizing comparators are given in Appendix C.

TABLE IV. Frozen structural constants and absolute held-out means. m is the control-register size, $|\mathcal{B}|$ the size of the exact-order success set, and the ideal column is the noiseless exact-order probability. Reference and intrinsic columns are means over the 12 meta-repetitions of that instance.

Instance	m	$ \mathcal{B} $	Ideal	Reference	Intrinsic
(15, 2)	4	2	0.500000	0.484375	0.485070
(15, 7)	4	2	0.500000	0.483979	0.484743
(21, 2)	7	2	0.228073	0.218434	0.218628
(21, 5)	7	2	0.228073	0.218673	0.219403
(33, 2)	7	2	0.175233	0.168084	0.168260
(35, 2)	8	4	0.228073	0.218959	0.219362

C. Response and multiplier preservation

Every intrinsic iterate is $\theta = Tz$ with $RT = 0$, so $R\theta = 0$ and $U(Tz) = U_0$ hold by construction within the declared ansatz. Executing the code verifies this numerically: over all six instances and all 72 meta-repetitions,

$$\max \|R\theta\| \approx 2.7 \times 10^{-16},$$

$$\max \|U(\theta) - U_0\| \approx 4.8 \times 10^{-16},$$

i.e. double-precision round-off. Because exact preservation follows analytically from the parameterization $\theta = Tz$ within the declared ansatz, these residuals are implementation audits of the executed numerical pipeline rather than independent physical evidence. They verify that the recorded improvement was not produced by a numerical implementation error that moved the reported iterates off the declared fibre; they say nothing about invariance on a physical device.

VI. MECHANISM DIAGNOSTICS AND ABLATIONS

The aggregate comparison establishes that intrinsic navigation outperforms the selected equal-budget baselines, but it does not identify why. We therefore perform exploratory diagnostics on the (15, 2) landscape, using 256 frozen tangent-space samples at the reference point. None of the following is offered as an identification of the source of the advantage.

A. Local curvature anisotropy

The tangent-space Hessian, estimated by central differences with step $h = 2 \times 10^{-3}$ in the orthonormal coordinates z , has eigenvalues

$$\lambda \approx [-0.0518, -0.0112, -0.00036, -0.00024],$$

giving the anisotropy ratio $\max_k |\lambda_k| / \min_k |\lambda_k| \approx 219.8$. The sampled landscape is therefore strongly anisotropic

near the reference. Negative Hessian eigenvalues describe local curvature but do not by themselves make the reference a local maximum, because the gradient at the reference need not vanish; the same frozen diagnostic records a projected training-gradient norm of $\|\nabla_z \hat{S}_{\text{train}}(0)\| \approx 6.9 \times 10^{-3}$ there, so the reference is not a critical point of the training objective. No step-size sensitivity analysis of h was performed. This is a local property of one instance: the diagnostic was not repeated on the five additional instances, and it does not establish that anisotropy is the source of the intrinsic method's advantage. No claim is made that the landscape is globally smooth or free of local optima.

B. Gate infidelity as an insufficient task surrogate

Over the same 256 samples, the correlation between a gate-infidelity proxy and task success is Pearson $\rho \approx -0.30$ and Spearman $\rho \approx -0.47$. The association is negative but moderate, so gate infidelity is not a sufficient surrogate for exact-order recovery within this landscape. This supports treating the objective as task-level rather than gate-level, but says nothing about the mechanism of the advantage.

C. Hessian-preconditioning ablation

A Hessian-preconditioned variant was tested as an exploratory ablation on (15, 2) and did not outperform the primary intrinsic Euclidean ascent ($\bar{S}_{\text{held,pre}} = 0.484996$ versus $\bar{S}_{\text{held,intrinsic}} = 0.485031$). The Hessian is accordingly treated as a diagnostic of local landscape geometry, not as a component of the reported method.

VII. DISCUSSION AND LIMITATIONS

The observed gain is not a change in the ideal Shor computation. It is a change in the implementation selected from a family that is indistinguishable at the declared ideal-response level: every compared implementation realizes the same U_0 , and the differences appear only once the finite-noise channel and the continued-fraction success criterion are applied. The effect is therefore task-level rather than gate-level, a reading reinforced by the weak correlation between the gate-infidelity proxy and exact-order recovery.

The existence of fibre-dependent task variation is built into the declared synthetic noise map; the empirical question addressed here is whether that variation can be navigated in a way that transfers from training seeds to shifted held-out seeds and remains directionally consistent across the frozen instances. The answer obtained is affirmative for the three specified comparator configurations at the stated hyperparameters, and for no wider class. The appropriate reading of this paper is therefore

a controlled proof of concept: exact ideal-response redundancy can support held-out task-level navigation in a stipulated synthetic model. It is not a demonstration that Shor’s algorithm is improved, that order finding is optimized in general, or that any advantage exists at the level of native circuits or hardware.

The following limitations bound the claim:

- Six small state-vector instances, all sharing one synthetic implementation ansatz and one noise-model family; they are not independent tests of the underlying effect, and the hierarchical bootstrap quantifies sampling variation across instances, not model misspecification.
- The ansatz is a synthetic nine-parameter phase parameterization, not a native gate decomposition; the mapping to a compiled circuit is not established.
- No hardware result, no fault-tolerance result, no cryptographic-scale claim, and no asymptotic Shor advantage.
- The absolute effect sizes are of order 10^{-4} in success probability, and no claim is made about the practical relevance of that magnitude.
- Full factoring involves many such blocks plus classical post-processing.
- The comparison is against three specified configurations. Random-search radius sensitivity, an SPSA hyperparameter sweep, multi-start optimization, multiplicity-adjusted or small-cluster-corrected intervals, and an independent clean-room reimplementations are outstanding and are listed as future work rather than performed here.

The natural next steps are a native compiled-circuit realization of the implementation freedom, a hardware-calibrated noise family in place of the synthetic one, and an independent clean-room replication of the frozen protocol.

VIII. CONCLUSION

We have shown that, within the declared synthetic ansatz and noise family, realizations of the same ideal Shor modular multiplier can exhibit different finite-noise exact-order recovery. At the stated hyperparameters, intrinsic Euclidean ascent on the four-dimensional exact-response fibre achieves higher held-out performance than

the unchanged reference and the two specified equal-budget fibre-restricted comparator configurations, with positive mean advantage on every frozen instance, while preserving the declared response and ideal multiplier to approximately 10^{-16} . The result is a finite-dimensional task-level example of navigable exact-response redundancy within a stipulated model. Whether this structure survives native compilation, hardware-calibrated noise, and larger instances remains open.

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CODE AND DATA AVAILABILITY

The v0.9.3 preflight bundle of Sec. IV — driver script, frozen protocol JSON, complete per-meta records, reference stdout, and SHA-256 manifest — is archived at <https://github.com/papasop/geometric-flow-qc> under tag `shor-task-fibre-v0.9.3`, commit `b3d38ba73376c14639618d8fb7f73eedd4c2c55b` (2026-08-16). The driver has SHA-256

```
ff91b6a8fde7228b40397df5aa24c218
fe5dfdcdb0e19ff68bb794a3eed55e2d4
```

and re-executing it under NumPy reproduces the archived result file bitwise, SHA-256

```
43bfa6e8da19409311f46e2dd93bd28e
c966bf01f041192cbe7db5c017118519
```

The v0.9.4 multi-instance driver, protocol JSON, complete per-meta records and reference stdout that underlie Sec. V are archived at <https://github.com/papasop/geometric-flow-qc> under tag `shor-multinstance-fibre-v0.9.4`, commit `24da8d7ea9dddb702eb07de4b567bf5637f01e51` (2026-08-17); the frozen driver has SHA-256

```
b005753cb4780f1d1aee2da463abb2f
61785ba5d27bf0ddc6d9db3e97bd43aac
```

The v0.9.4 protocol and driver were frozen locally before execution of the full multi-instance run and were archived publicly after completion. This was not a registry-based preregistration, and no claim of prior public registration is made.

Manuscript revisions made after the frozen run affect exposition, disclosure, and presentation only; the v0.9.4 code, protocol, archived records, and numerical results are unchanged. The present manuscript revision is designated v0.9.4-m3; the computational protocol remains v0.9.4, and no new computational tag was created.

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Appendix A: Complete Response Matrix

The frozen response matrix, identical for all six instances, is

$$R = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & -1 & 0 & 1 & 1 & 1 & 1 \end{bmatrix},$$

with $\text{rank}(R) = 5$. The sixth row is the sum of rows 1, 2, 3, and 5 and is retained as an explicit consistency/audit coordinate. Rows 1–5 are independent, so $\text{rank}(R) = 5$ and $\dim \ker R = 4$ as stated in Sec. II. In the code the projector is constructed via the Moore-Penrose pseudoinverse,

$$P_T = I_9 - R^\dagger R,$$

and T is an orthonormal basis of the range of P_T obtained by a frozen deterministic factorization.

Appendix B: Complete Noise Model

The synthetic noise model is defined below. All random number generation uses NumPy’s default generator with the seeds specified in the frozen protocol JSON.

1. Common calibration matrix

All instance-level seeds are produced by one integer mixing rule, chosen so that they are independent of Python hash randomization:

$$\sigma(N, a, \text{salt}) = (20260816 + 1000003 N + 9176 a + 7919 \text{salt}) \bmod 2^{32}.$$

For each instance a fixed common calibration matrix $C_0 \in \mathbb{R}^{m \times 9}$ is drawn once from the generator seeded with $\sigma(N, a, 17)$, and each row is normalized to unit norm,

$$(C_0)_{k,:} \leftarrow \frac{(C_0)_{k,:}}{\|(C_0)_{k,:}\|_2}.$$

2. Per-noise-seed drift

For every noise seed ξ an independent drift matrix $D_\xi \in \mathbb{R}^{m \times 9}$ is drawn with i.i.d. standard normal entries and normalized row-wise. The effective calibration matrix is

$$C_\xi = \text{rownorm}(C_0 + \kappa D_\xi),$$

with $\kappa = 0.10$ for training seeds and $\kappa = 0.17$ for shifted held-out seeds.

3. Phase offsets

For a control register state $x \in \{0, \dots, M-1\}$ define the binary digits

$$b_k(x) = \left\lfloor \frac{x}{2^k} \right\rfloor \bmod 2, \quad k = 0, \dots, m-1.$$

The phase offset is

$$\phi_x(\theta, \xi) = \sum_{k=0}^{m-1} b_k(x) \epsilon_k(\theta, \xi),$$

where

$$\epsilon(\theta, \xi) = \epsilon^{\text{common}} + \delta_\xi + 0.72 C_\xi \theta,$$

and $\epsilon^{\text{common}} \in \mathbb{R}^m$ is the frozen common phase vector, generated for every control-register size by the single deterministic rule

$$\epsilon_j^{\text{common}} = 0.105 (-0.78)^j, \quad j = 0, \dots, m-1.$$

The same extension rule therefore applies to all six instances; its first four entries are $[0.105, -0.0819, 0.063882, -0.04982796]$. The per-seed phase perturbation is

$$(\delta_\xi)_k \sim \text{i.i.d. } \mathcal{N}(0, \sigma^2),$$

with $\sigma = 0.010$ for training seeds and $\sigma = 0.016$ for shifted held-out seeds.

4. Dephasing

The dephasing rate is drawn independently per seed,

$$\eta_\xi = \text{clip}_{[0,0.15]}(\mu_\eta + \zeta_\xi), \quad \zeta_\xi \sim \mathcal{N}(0, 0.004^2),$$

with $\mu_\eta = 0.025$ for training seeds and $\mu_\eta = 0.040$ for shifted held-out seeds. Held-out seeds are therefore shifted in three independent ways: calibration drift, phase noise scale, and dephasing rate.

5. Seed hierarchy

The j -th meta-repetition of an instance uses the seed $\sigma(N, a, 101 + 1009j)$ for $j = 0, \dots, 11$. A single deterministic generator is initialized from it, which first draws the five training seeds and then the eight held-out seeds sequentially, each as a 31-bit integer. The comparator algorithms draw their own randomness from generators seeded with the meta seed XOR $\sigma(N, a, 31)$ for random search and XOR $\sigma(N, a, 53)$ for SPSA, so comparator randomness is independent of the noise seeds. Training and held-out seed sets are disjoint by construction, and the seed hierarchy is instance-specific, so the (15, 2) seeds of v0.9.4 differ from those of the v0.9.3 preflight.

Appendix C: Statistical Methods

TABLE V. Absolute held-out means of the two comparators, each over the 12 meta-repetitions of that instance.

Instance	Random	SPSA
(15, 2)	0.484713	0.484867
(15, 7)	0.484451	0.484675
(21, 2)	0.218397	0.218567
(21, 5)	0.218826	0.219133
(33, 2)	0.168075	0.168196
(35, 2)	0.218957	0.219306

The aggregate intervals are equal-instance-weight hierarchical bootstrap percentile intervals: for each of 10,000 draws, the six instances are resampled with replacement, then the 12 meta-repetitions are resampled with replacement within each sampled instance, and the equal-weight mean over sampled instances is recorded; the reported interval spans the 2.5% and 97.5% quantiles. The per-instance entries of Table III are plain means over the 12 meta-repetitions of that instance.

The post hoc sign test reported in Sec. V is a two-sided exact binomial test over the six instance-level mean differences of the v0.9.4 multi-instance study. It was not part of the frozen primary decision rule.

All comparisons are made on $\hat{S}_{\text{held}}$, never on the training objective used for optimization.

Table V completes the absolute scale of Table IV by giving the held-out means of the two optimizing comparators.